



Kinetics of hydrolysis of poly(ethylene terephthalate) wastes catalyzed by dual functional phase transfer catalyst: A mechanism of chain-end scission

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ABSTRACT

The dual functional phase transfer catalyst $[(\text{CH}_3)_3\text{N}(\text{C}_{16}\text{H}_{33})]_3[\text{PW}_{12}\text{O}_{40}]$ was used in the hydrolysis of poly(ethylene terephthalate) (PET) wastes and exhibited outstanding catalytic activity. In this work, the kinetic study about this reaction was carried out. The quantification of ethylene glycol (EG) formed in the hydrolysis was used to describe the reaction rate, which could result in more feasible and accurate kinetic analysis than terephthalic acid (TPA) formation or PET consumption used in traditional method. The results of kinetic analysis showed that the depolymerization reaction rate was first order to PET and catalyst concentration. This indicated that the ester linkages on the surface of the PET sequentially reacted with water to produce products. A mechanism for the reaction occurring on the polymer chain end section was proposed. Also the activation energy value was derived. Kinetic correlations provided a reliable mathematical reaction model for this recycling process, which is in agreement with the principles of sustainable development.

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1. Introduction

Poly(ethylene terephthalate) (PET) exhibits excellent characteristic features, and with the rapid increase in consumption of PET, the amount of waste PET is growing rapidly [1]. PET does not generate a direct harm to the environment, but it is seen as a noxious material, hence its substantial fraction by volume in the waste and its high resistance to the atmospheric [2]. Therefore, PET recycling represents one of the most widespread examples of polymer recycling. The recycling of waste polymers including PET can be carried out in many ways [3–15]. Recently, there is a growing interest in hydrolysis for the chemical recycling of PET, since it leads to the formation of the monomers of terephthalic acid and ethylene glycol, from which the polymer is made of.

Our group has reported the hydrolysis of PET in the presence of dual functional phase transfer catalyst [16]. The dual functional phase transfer catalyst could behave as the efficient catalyst for the depolymerisation of PET. The excellent conversion of PET and yield of recovered product terephthalic acid (TPA) were obtained. Investigation into recycling of the catalyst showed that the catalyst could be reused without significant decrease of activity at least three cycles. This process provided an economically and convenient approach for the hydrolysis of PET waste bottles. In this work, based on our continuing interest in the depolymerisation of polyester PET, we carried out the comprehensively kinetic studies in order to further understand the process of dual functional phase transfer catalyzed hydrolysis. In spite of the widely proved feasibility, taking into account social, economic and environmental aspects of PET wasters by hydrolysis, relatively few works have been devoted to the kinetic description of PET hydrolysis [17–21]. Most published reports studied

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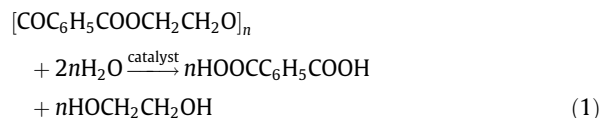
the kinetics by estimating the yield of TPA or the consumption of PET, however, this procedure involved a significant experimental error. The research neglected one important issue that the TPA recovered process included many steps which might result in the loss of product and in turn may affect kinetic calculations. Therefore, in this work three sets of data including TPA yield, ethylene glycol (EG) yield and PET conversion were recorded and compared to declare and afford accurate results. The specific objective of this study is not only to determine the kinetics of the catalyzed reaction in a batch reactor with dual functional phase transfer catalyst as catalyst, but also to provide a new pathway to study the kinetics in lab or under scale-up condition. Hence, integral kinetic analysis of the data was performed in order to establish the kinetic model and the kinetic parameters, which are necessary for developing the industrial process of depolymerisation of PET.

2. Experimental

PET flakes, dual functional phase transfer catalyst and water were charged into the reactor at room temperature and then heated to the selected temperature (115–145 °C) for conducting the hydrolytic runs. The mixture was allowed to react for some hours. Both temperature and pressure were kept constant during each experiment. In all kinetic studies, reaction time zero was taken to be the time at which the reactor temperature was the predetermined reaction temperature. After the required time interval for reaction was reached, the vessel was quickly removed from the heating mantle and immersed in an ice bath. The temperature of the vessel was quenched to room conditions so as to interrupt the progress of hydrolysis.

The synthesis of dual functional phase transfer catalyst $[(CH_3)_3N(C_{16}H_{33})]_3[PW_{12}O_{40}]$, workup of hydrolysis and calculating of the conversion and yield were carried out following the procedure described in previous paper published by our group [16]. The ethylene glycol (EG) content was determined with the GC analysis by working curve method. Conversion of PET was calculated by following formula: Conversion of PET = $[(w_1 - w_2)/w_1] \times 100\%$. Yield of TPA = $(w_3/w) \times 100\%$, where w_1 is the original PET weight, w_2 is the undepolymerized PET weight, w_3 is the TPA product weight, w is the weight of TPA which should be obtained theoretically.

3. Results and discussion



PET flakes were hydrolyzed with water catalyzed by dual functional phase transfer catalyst to yield the TPA and EG [16]. The reaction of Eq. (1) was clean and high-yielding reaction, and the purity of recovered TPA was checked and analyzed by means of a number of analytical techniques (1H NMR, FTIR and elemental analysis) and the results were the same as those corresponding to a commercial sample. Therefore, kinetic studies were carried

out to evaluate the reaction orders, rate constants and activation parameters for mechanistic understandings. In our published results, the dual functional phase transfer catalyst $[(CH_3)_3N(C_{16}H_{33})]_3[PW_{12}O_{40}]$ exhibited the best catalytic activity, therefore, an attempt to develop a kinetic model was made with it as a catalyst in this work.

A set of kinetic experiments were conducted in the 115–145 °C temperature range and varying PET: catalyst molar ratio 300:1 and 100:1. The regular depolymerisation and random scissions were two main forms of polymer degradation. The results included in Fig. 1 showed that the TPA yield was proportional with reaction time. The yield at 145 °C lower than 100% might be due to the uncompleted decomposition of PET. No depolymerisation conversion jump was observed, which suggests that the reaction occurring on the polymer chain end section should be the major reaction. It is proposed as a typical regular depolymerisation. This finding was in stark contrast with the hydrolysis mechanism observed for the microwave assisted hydrolysis [22]. The different heating method and reaction temperature might attribute to the huge difference of reaction mechanism. PET degradation into TPA was favoured with increasing temperature and catalyst concentration. The influence of temperature on yield was larger than that of the catalyst concentration. On the other hand, the molecular weights of unreacted flakes after reaction were measured. And the results showed that the molecular weight of remaining PET flakes was slightly decreased with an increase of the reaction temperatures and no significant change was observed. Therefore, the bulk phase of remaining flakes after the reactions was close to the fresh long chain PET. This suggested that the depolymerization of PET catalyzed by the dual functional phase transfer catalyst occurred on the external surface of PET flakes, and the flakes were lamellarly depolymerized.

Most reports regarding the hydrolysis of PET estimate the yield of TPA or the consumption of PET, and consequently the conversion of the process from the amount of TPA recovered after filtration followed by titration and dry. However, this procedure involves a significant experimental error, since the TPA recovered process included many steps which might result in the loss of product and in turn may affect kinetic calculations. In contrast, a quantification of EG formed after hydrolysis by GC can lead to the more accurate kinetic data. Fig. 2 shows the comparison of conversion values calculated by three methods. Data included in this figure were recorded from numerous reaction runs performed under varying operation conditions in terms of temperature, reaction time and concentration of catalyst. From the figure, good correlation was found between the yields of EG after hydrolysis with PET conversions, however, the TPA yields were slightly lower than those of EG and PET conversions under the reaction conditions described in this work. The result indicated that a minor quantity of TPA would unavoidably not be recoverable in this system. Moreover, it must be highlighted that the analytical method of EG formed would be likely to be employed under real conditions after a scale-up for the easy dealing process. Likewise, quantification of EG was more feasible and accessible than measurement of

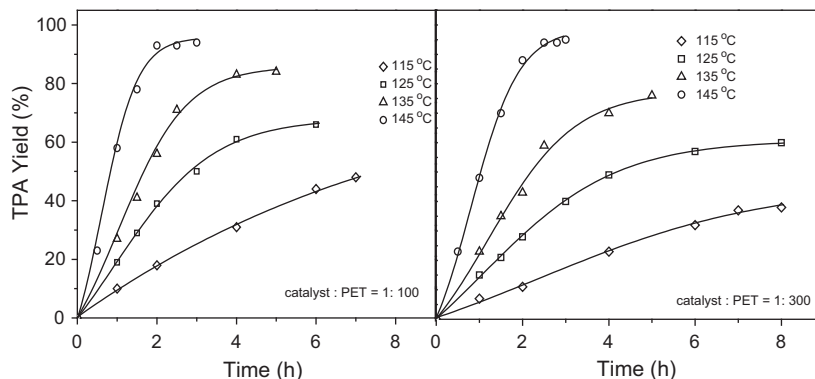
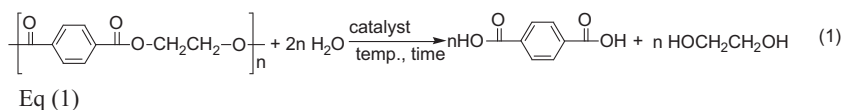


Fig. 1. TPA yield as a function of reaction time with varying catalyst concentration at several reaction temperatures.

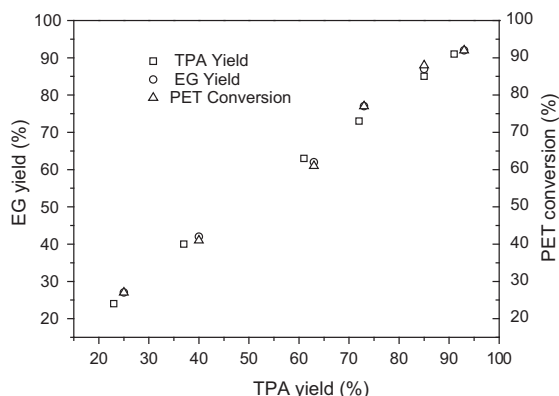


Fig. 2. Comparison of TPA yield, EG yield and PET conversion.

consumption of PET while scale-up. On the other hand, good accordance between EG yield with PET conversion proved that the solid PET depolymerisation produced almost only TPA and EG catalyzed with dual functional phase transfer catalyst. Thus, the change of EG concentration with hydrolysis could be selected as a function to record the reaction process. The following kinetic studies were based on the concentration of EG estimated after hydrolysis.

The reaction of PET with water under acidic condition was proposed to involve depolymerisation and reversible esterification by degradation products. The reaction rate of PET hydrolysis was defined by the molar consumption of PET (or molar formation of TPA or EG) with reaction time. Therefore, the PET catalytic depolymerisation rate was defined according to the following equation,

$$\text{Rate} = k_1 C_{\text{PET}}^a C_{\text{H}_2\text{O}}^b C_{\text{CAT}}^c - k_2 C_{\text{CAT}}^c C_{\text{TPA}}^d C_{\text{EG}}^e \quad (2)$$

where a , b , c , d and e are the reaction orders with respect to PET, H_2O , catalyst, TPA and EG, respectively, k_1 is the rate constant of the direct reaction and k_2 is the rate constant

of the inverse reaction. C_{PET} , $C_{\text{H}_2\text{O}}$, C_{CAT} , C_{TPA} and C_{EG} represent the molar concentration of PET, H_2O , catalyst, TPA and EG, respectively.

By setting $C_{\text{H}_2\text{O}}^{\text{initial}} \gg C_{\text{PET}}^{\text{initial}}$ (real H_2O :PET molar ratio is 15:1), it could be assumed that H_2O was in excess in the reaction mixture and therefore its concentration remained virtually constant during the reaction. On the other hand, the TPA and EG were produced as the reaction proceeded, which were monomers in an eventual nucleophilic substitution for the reverse reaction for the hydrolysis. The reaction of TPA and EG were examined and proved to be nonreactive under the same reaction condition for the hydrolysis in the presence of the dual functional phase transfer catalyst. Therefore, the reaction of PET hydrolysis under the mentioned conditions could be considered as an irreversible reaction. Moreover, as a first approach it was assumed that the reaction rate was first order with respect to the concentration of all species involved in the reaction. The overall rate of the process could be simply formulated as the following power form in Eqs. (3)–(5), thus establishing pseudo-first-order kinetics. The quantity of EG formed after hydrolysis was select-

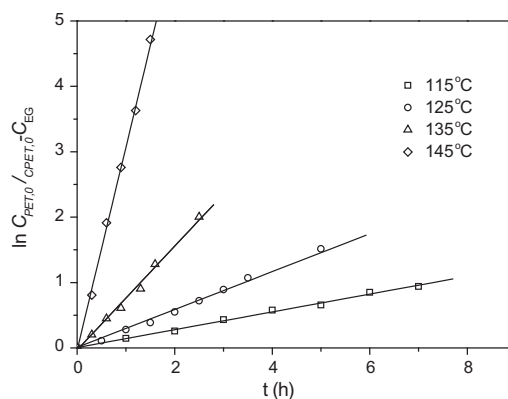


Fig. 3. Fitting of kinetic data according to Eq. (8) at different temperatures (PET:catalyst molar ratio: 1:100).

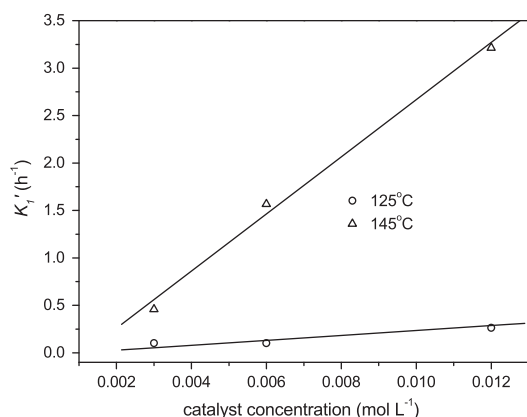


Fig. 4. Linearity of k'_1 vs C_{CAT} .

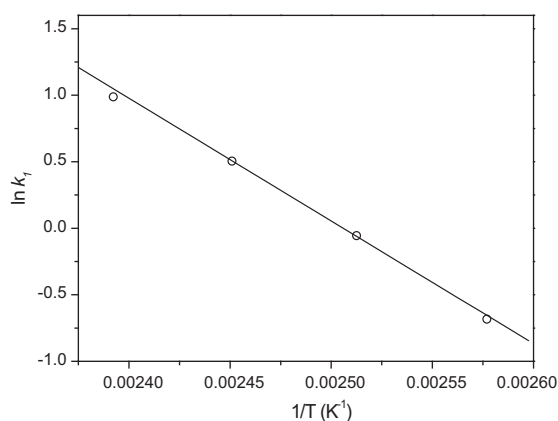


Fig. 5. Arrhenius plot of the k_1 in the presence of dual functional phase transfer catalyst.

ed to be a function data to establish the kinetics. The rate law equation was rewritten as Eq. (6) using formation rate of EG to express the hydrolysis rate. Then the Eq. (7) was substituted into Eq. (6) and integrated to derive the following kinetic expression Eq. (8).

$$\text{Rate} = k_1 C_{PET} C_{H_2O} C_{CAT} \quad (3)$$

$$= k'_1 C_{PET} \quad (4)$$

$$k'_1 = k_1 C_{H_2O} C_{CAT} \quad (5)$$

$$dC_{EG}/dt = k'_1 C_{PET} \quad (6)$$

$$C_{PET} = C_{PET,0} - C_{EG} \quad (7)$$

$$\ln C_{PET,0}/C_{PET,0} - C_{EG} = k'_1 t \quad (8)$$

After examining the kinetic model by fitting the data recorded at different temperatures, a typical curve was shown in Fig. 3 and pseudo first-order rate constant k'_1 were derived from the slopes of such plots. Results indicated that the model provided a good fit for our experimental data. Therefore, it was concluded that the reaction rate was first order with respect to the PET concentration. Moreover, the linear relationship between the rate constant k'_1 and the catalyst concentration shown in Fig. 4 confirmed that the reaction rate was also first order with respect to the catalyst concentration. Further, the reaction rate k_1 at the selected temperatures related to the catalyzed reaction could be estimated from the slopes of the plots. On the other hand, the intercept of this line would be related to the rate constant of the uncatalyzed reaction. From the figure, it can be seen that the intercept was almost equal to zero. This result agreed with the real experimental result, no PET depolymerization occurring at the 125 °C and 145 °C without catalyst added. Thereby, the negligible contribution of the non-catalytic reaction rate to the overall process was confirmed. Hence, results clearly evidenced first order with respect to the PET and catalyst concentration, respectively, which suggesting that the ester chain in PET was hydrolyzed sequentially.

Arrhenius plot was shown in Fig. 5 for the determination of the activation parameters for the reaction of PET hydrolysis. Evaluation of the activation parameters over the temperature range 115–145 °C yielded activation energy of PET hydrolysis was 68 kJ mol⁻¹. The value estimated for the activation energy is lower than those for acid-hydrolysis, 100–110 kJ mol⁻¹ [18,23], neutral hydrolysis 90–123 kJ mol⁻¹ [24,25] and alkaline hydrolysis with phase transfer catalyst, 75–83 kJ mol⁻¹ [19,21]. So, the dual functional phase transfer catalyst indeed lowered the activation energy of the hydrolysis process.

From the kinetic studies, an interfacial mechanism was proposed for this hydrolysis of PET. The catalyst molecule could be uniformly distributed in the aqueous–organic interface and form a film. Consequently, the lipophilic quaternary ammonium cations would lie on the organic side and the hydrophilic heteropolyanions would lie on the aqueous side. The reaction mixture consists of the solid organic phase of the PET and the aqueous phase, so the dual functional phase transfer catalyst works with interfacial mechanism. The hydrolysis of the dual functional phase transfer catalyst occurred to afford the phase transfer catalyst and heteropoly acid. The phase transfer catalyst made the PET macromolecules of the surface of the flakes can easily be hydrolyzed. Meanwhile, the heteropoly acid

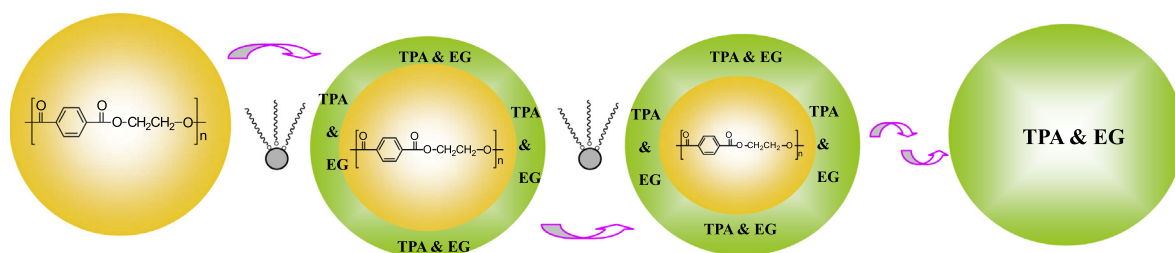


Fig. 6. Proposed interfacial mechanism.

worked as a highly active acidic catalyst. The reaction proceeds until complete depolymerisation of PET to TPA and EG, while the catalyst remains in the aqueous phase. The catalyst is regenerated in its added form and the catalytic cycle continues. As shown in Fig. 6, this reaction mechanism is different from that for the hydrolysis of PET either in neutral water or microwave assisted, where a random scission model contributes to the breaking of ester chain and subsequently produces the products [3,20,26,27]. Nevertheless, the PET depolymerization reaction rate in the neutral water solution is much slower than that under the dual functional phase transfer catalyst. Therefore, high temperatures are required in order to enhance the reaction rate, in contrast to those below 145 °C for the reaction in this work. Likewise, the proposed mechanism was consistent with the experimental data summarized in Fig. 1.

4. Conclusions

Dual functional phase transfer catalyzed PET hydrolysis taken from post-consumer soft-drink bottles was revealed to be an efficient method for the degradation of PET and reproduction of pure terephthalic acid. The correlation between the yields of products and the conversion of PET showed that the major hydrolysis of PET linkages was through a mechanism of chain-end scission. The kinetics of the depolymerization reaction was extensively studied. A simple theoretical model was developed to describe the hydrolysis rate. The result of kinetics showed that the reaction rate was first order each to the concentration of PET and to the catalyst, respectively. This suggests that the ester chain in PET was hydrolyzed sequentially. The activation energy for the PET hydrolysis catalyzed by dual functional phase transfer catalyst was calculated as 68 kJ mol⁻¹.

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References

- [1] Yoshioka T, Ota M, Okuwaki A. Conversion of a used poly(ethylene terephthalate) bottle into oxalic acid and terephthalic acid by oxygen oxidation in alkaline solutions at elevated temperatures. *Ind Eng Chem Res* 2003;42:675–9.
- [2] Scheirs J. Chapter No 4: Recycling of PET. In: *Polymer recycling*. Wiley series in polymer science. Sussex: J. Wiley & Sons; 1998.
- [3] Sato O, Arai K, Shirai M. Hydrolysis of poly(ethylene terephthalate) and poly(ethylene 2,6-naphthalene dicarboxylate) using water at high temperature: effect of proton on low ethylene glycol yield. *Catal Today* 2006;111:297–301.
- [4] deCarvalho GM, Muniz EC, Rubira AF. Hydrolysis of post-consumer poly(ethylene terephthalate) with sulfuric acid and product characterization by WAXD, ¹³C NMR and DSC. *Polym Degrad Stab* 2006;91:1326–32.
- [5] Guclu G, Yaicinyuva T, Ozgumus S, Orbay M. Hydrolysis of waste PET and characterization of products by differential scanning calorimetry. *Thermochim Acta* 2003;404:193–205.
- [6] Grause G, Kaminsky W, Fahrback G. Hydrolysis of poly(ethylene terephthalate) in a utilized bed reactor. *Polym Degrad Stab* 2004;85:571–5.
- [7] Firas A, Dumitru P. Recycling of PET. *Eur Polym J* 2005;41:1453–77.
- [8] Karayannidis GP, Achilias DS, Sideridou ID, Bikiaris DN. Alkyd resins derived from glycolized waste poly(ethylene terephthalate). *Eur Polym J* 2005;41:201–10.
- [9] López-Fonseca R, Duque-Ingunza I, de Rivas BS, Arnaiz JI, Gutiérrez-Ortiz. Chemical recycling of post-consumer PET wastes by glycolysis in the presence of metal salts. *Polym Degrad Stab* 2010;95:1022–8.
- [10] Abdelaal MY, Sobahi TR, Makki MSI. Chemical transformation of pet waste through glycolysis. *Constr Build Mater* 2011;25:3267–71.
- [11] Viana ME, Riul A, Carvalho GM, Rubira AF, Muniz EC. Chemical recycling of PET by catalyzed glycolysis: kinetics of the heterogeneous reaction. *Chem Eng J* 2011;173:210–9.
- [12] Shukla SR, Harad AM. Aminolysis of polyethylene terephthalate waste. *Polym Degrad Stab* 2006;91:1850–4.
- [13] Pingale ND, Palekar VS, Shukla SR. Glycolysis of postconsumer polyethylene terephthalate waste. *J Appl Polym Sci* 2010;115:249–54.
- [14] Inha VS, Patel MR, Patel JV. PET waste management by chemical recycling “a review”. *J Polym Environ* 2010;18:8–25.
- [15] Mittal A, Soni RK, Dutt K, Singh S. Scanning electron microscopic study of hazardous waste cakes of polyethylene terephthalate (PET) by aminolysis and ammonolysis. *J Hazard Mater* 2010;178:390–6.
- [16] Zhang LR, Gao J, Zou JZ, Yi FP. Hydrolysis of poly(ethylene terephthalate) waste bottles in the presence of dual functional phase transfer catalysts. *J Appl Polym Sci* 2013;130:2790–5.
- [17] Wan BZ, Kao CY, Cheng WH. Kinetics of depolymerization of poly(ethylene terephthalate) in a potassium hydroxide solution. *Ind Eng Chem Res* 2001;40:509–14.
- [18] Yoshioka T, Okayama N, Okuwaki A. Kinetics of hydrolysis of poly(ethylene terephthalate) powder in sulfuric acid by a modified shrinking-core model. *Ind Eng Chem Res* 2001;40:75–9.
- [19] Lopez-Fonseca R, Gonzalez-Marcos MP, Gonzalez-Velasco JR, Gutierrez-Ortiz JI. A kinetic study of the depolymerization of poly(ethylene terephthalate) by phase transfer catalysed alkaline hydrolysis. *J Chem Technol Biotechnol* 2009;84:92–9.
- [20] Campanelli JR, Kamal MR, Cooper DG. A kinetic study of the hydrolytic degradation of polyethylene terephthalate at high temperatures. *J Appl Polym Sci* 1993;48:443–51.
- [21] Kosmidis V, Achilias DS, Karayannidis G. Poly(ethylene terephthalate) recycling and recovery of pure terephthalic acid, kinetics of phase transfer catalyzed alkaline hydrolysis. *Macromol Mater Eng* 2001;286:640–7.
- [22] Tsintzou GP, Antonakou EV, Achilias DS. Environmentally friendly chemical recycling of poly(bisphenol-A carbonate) through phase transfer-catalysed alkaline hydrolysis under microwave irradiation. *J Hazard Mater* 2012;241–242:137–45.
- [23] Mancini S, Donnini ZM. Post consumer pet depolymerization by acid hydrolysis. *Polym Plast Technol Eng* 2007;46:135–44.
- [24] Kao CY, Wan BZ, Cheng WH. Kinetics of hydrolytic depolymerization of melt poly(ethylene terephthalate). *Ind Eng Chem Res* 1998;37:1228–34.
- [25] Goje AS, Thakur SA, Diware VR, Patil SA, Dalwale PS, Mishra S. Chemical recycling, kinetics, and thermodynamics of hydrolysis of poly(ethylene terephthalate) waste with nonaqueous potassium hydroxide solution. *Polym Plast Technol Eng* 2004;43:369–88.
- [26] Campanelli JR, Kamal MR, Cooper DG. Catalyzed hydrolysis of polyethylene terephthalate melts. *J Appl Polym Sci* 1994;53:985–91.
- [27] Launay A, Thominette EV. Hydrolysis of poly(ethylene terephthalate). A steric exclusion chromatography study. *Polym Deg Stab* 1999;63:385–9.